

Incident Response

Condensed Study Notes

Generated: 2026-05-24 23:40 UTC

WEKK 5 INCIDENCE RESPONSE

This section introduces the concept of 'incidence response' within the context of WEKK 5.

An 'incidence' refers to an event that disrupts normal operations or poses a threat. The 'response' to such an incidence is the set of actions taken to manage and mitigate the event.

Think of it like a fire alarm going off in a building. The alarm itself is the incidence. The response would be the actions taken by people in the building and emergency services to ensure safety and put out the fire. This includes evacuating, calling the fire department, and the fire department's efforts to extinguish the fire and assess the damage.

Key aspects of incidence response typically involve:

- **Detection:** Identifying that an incidence has occurred.
- **Containment:** Limiting the spread or impact of the incidence.
- **Eradication:** Removing the cause of the incidence.
- **Recovery:** Restoring systems or operations to their normal state.
- **Post-incidence Analysis:** Learning from the event to improve future responses.

Introduction

Incident response is a structured and strategic method for dealing with cyber attacks. Its main goals are to minimize the damage caused by an attack, reduce the time it takes to recover, and lower the overall costs involved.

Think of incident response as a specialized team within a larger organization. Incident management is the broader organizational effort to handle cyber attacks, involving many different departments like executives, legal, HR, communications, and IT. Incident response, on the other hand, focuses specifically on the technical cybersecurity tasks and considerations within that larger management framework.

Types of security incidents

To create effective incident response strategies, it's helpful to understand how vulnerabilities, threats, and incidents are connected.

- **Vulnerability:** A weakness in an IT system or business environment. Think of it like an unlocked door in a house.
- **Threat:** An entity (like a hacker or an insider) that aims to exploit a vulnerability. This is like someone noticing the unlocked door and intending to go inside.
- **Incident:** An attack that successfully accesses enterprise resources or puts them at risk. This is when the person with bad intentions actually enters the house through the unlocked door.

- **Data Breach:** A specific type of incident where attackers successfully compromise sensitive information, such as personal details or company secrets. This is like the intruder stealing valuables from the house.

Organizations should proactively fix vulnerabilities and prepare for common security incidents. These incidents include:

- **Privilege Escalation Attacks:** Gaining higher-level access than initially authorized.
- **Man-in-the-Middle (MitM) Attacks:** Intercepting communication between two parties to eavesdrop or alter messages.
- **Malware Attacks:** Using malicious software (like viruses or ransomware) to disrupt systems or steal data.
- **Unauthorized Access Attempts:** Trying to get into systems or data without permission.
- **Advanced Persistent Threats (APTs):** Sophisticated, long-term attacks often by well-funded groups, aiming to gain and maintain access to a network.
- **Web Application Attacks:** Exploiting weaknesses in websites or web applications.
- **Insider Threats:** Malicious actions taken by individuals within an organization (employees, contractors, etc.).
- **Phishing Attacks:** Deceptive attempts to trick individuals into revealing sensitive information, often through fake emails or websites.
- **Password Attacks:** Attempts to guess or crack user passwords to gain unauthorized access.
- **Denial-of-Service (DoS) Attacks:** Overwhelming a system with traffic to make it unavailable to legitimate users.

How to create an incident response plan

Creating a strong incident response plan involves careful preparation before any disruptive event occurs. This means drafting, reviewing, and testing your plans in advance.

A good incident response plan acts like a guide, empowering the people who handle incidents and helping them make smart decisions when things get tough.

Here's how to build one:

1. **Establish a Policy:** This is a high-level document that outlines the general priorities for handling security incidents. Think of it as the constitution for incident response – setting the overarching goals and principles.
2. **Build an Incident Response Team:** The effectiveness of your plan depends on the people executing it. You need to define who is responsible for what tasks and ensure everyone on the team has the necessary training to do their job well. This team is the core group that will spring into action.
3. **Create Playbooks:** While the policy is high-level, playbooks are where the detailed actions are laid out. These are standardized, step-by-step guides for responding to specific types of incidents. Playbooks are crucial for ensuring consistency, efficiency, and effectiveness, and they are also excellent tools for training new responders. Imagine them as the detailed instruction manuals for different emergency scenarios.

4. Create a Communication Plan: An incident response plan needs clear communication to succeed. This plan details how and when to communicate with various stakeholders. These stakeholders can include internal teams (like IT, legal, HR, and executives), as well as external parties such as customers, partners, law enforcement, and the public.

What is an incident response plan

An incident response plan is essentially an organization's comprehensive guide for managing security incidents. Think of it like a detailed emergency preparedness manual for your digital world.

It acts as an authoritative map, directing responders through every stage of an incident, from the moment it's detected and assessed, all the way to containing and resolving it.

This plan specifically clarifies:

- **Who:** It identifies who is responsible for specific tasks during an incident and provides contact information for those individuals.
- **What:** It defines which threats, exploits, and situations are considered actionable security incidents and outlines the appropriate actions to take when they occur.
- **How:** It details the specific methods and procedures team members should follow to complete their assigned tasks.
- **When:** It specifies the circumstances under which particular tasks should be performed by team members.

Incident response frameworks: Phases of incident response

Instead of building an incident response plan from scratch, organizations can use established frameworks for guidance. While frameworks from NIST, SANS Institute, ISO, and ISACA may have slight differences, they all outline similar key phases for managing security incidents.

These phases provide a structured approach to handling disruptive events:

- **Preparation/Planning:** This initial phase involves setting up the incident response team, developing policies and processes, creating detailed playbooks (step-by-step guides for specific scenarios), and deploying necessary tools and services. Think of this as stocking your emergency preparedness kit and writing down the evacuation plan before a storm hits.
- **Detection/Identification:** Here, IT monitoring systems are used to spot, evaluate, confirm, and prioritize potential security incidents. This is like having sensors and alarms that alert you when something unusual is happening.
- **Containment:** The goal is to stop the incident from spreading and to regain control of affected IT resources. This phase is similar to isolating a fire to prevent it from engulfing the entire building.
- **Eradication:** This involves completely removing the threat, such as malware or unauthorized user accounts, and identifying the vulnerabilities that allowed the attack to occur. It's like finding the source of the fire and putting it out, then patching the hole in the wall that let the wind in.
- **Recovery:** Once the threat is gone, the focus shifts to restoring normal operations and fixing any weaknesses that were exploited. This is the phase where you rebuild and reinforce what was damaged.

- **Lessons Learned:** After the incident is resolved, a thorough review takes place to understand what happened, when, and how. This phase is crucial for identifying any security controls, policies, or procedures that didn't work as well as they should have, and for updating the incident response plan to prevent future occurrences. It's like debriefing after a drill to see what went right and what needs improvement for the next time.

Types of incident response teams

Organizations establish specialized teams to handle security incidents. The raw content identifies three common types of incident response teams:

- **Computer Incident Response Team (CIRT)**
- **Computer Security Incident Response Team (CSIRT)**
- **Computer Emergency Response Team (CERT)**

While the raw content lists these as distinct types, it's important to note that these acronyms are often used interchangeably in the field, and the specific responsibilities and structures can vary greatly between organizations.

Why is an Incident Response Plan Important

An Incident Response Plan (IRP) is crucial for organizations for several key reasons:

- **Minimizing the Impact of Incidents:** A well-structured IRP allows organizations to react quickly and effectively to cyber threats. By following pre-defined steps, they can reduce downtime, financial losses, and damage to their reputation. Think of it like a fire escape plan: having one means you know exactly what to do when smoke appears, helping everyone get out safely and minimizing harm.
- **Improving Incident Detection:** An effective IRP incorporates continuous monitoring and regular reviews of security systems. This proactive approach helps organizations spot threats earlier, enabling them to take action before a full-blown breach occurs.
- **Streamlining Communication:** A core part of any IRP is ensuring clear communication among all involved parties, including IT staff, management, and employees. This keeps everyone informed and aligned during an incident, reducing confusion and speeding up recovery.
- **Legal and Regulatory Compliance:** Many industries have specific rules about data protection and cybersecurity. Having a documented IRP shows an organization's commitment to meeting these regulations and can help avoid penalties.
- **Building Resilience:** The process of creating and maintaining an IRP fosters a greater awareness of security among employees. When staff understand potential threats and how to respond, the organization as a whole becomes better equipped to handle cyberattacks.